DCAF: Differential Circuit Analysis Framework

Integrating Multi-Signal Training Perturbations, Projection-Level Granularity, and Triangulated Confidence for Neural Circuit Discovery

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Abstract

Mechanistic interpretability has produced powerful methods for discovering neural circuits, yet existing approaches typically rely on a single attribution technique, lack calibrated confidence scores, and operate at head-level or edge-level granularity—missing the projection-level structure (e.g., W_Q, W_K, W_V, W_O) where behavioral signatures diverge within the same component. We introduce the Differential Circuit Analysis Framework (DCAF), which treats training signals as controlled perturbation experiments: by training a model under 11 paired behavioral objectives (5 target-reinforcing, 5 opposite-reinforcing, 1 neutral baseline) and triangulating evidence across weight deltas, activation patterns, and latent geometry, DCAF discovers candidate circuit components, maps their causal connections, and classifies their functional roles. Key technical contributions include: (i) the 11-paired-objective probe configuration that uses preference optimization, supervised fine-tuning, gradient ascent, and sequential training as orthogonal discovery signals; (ii) three complementary discovery paths (weight-based, activation-based, gradient-based) integrated through a triangulated confidence score; (iii) the Direction Convergence Score (DCS), which measures cross-signal agreement on behavioral directions in the residual stream; and (iv) a 7-phase ablation protocol with interaction discovery and adaptive tiered functional classification. The framework is formulated in an architecture-agnostic manner with a complete transformer instantiation. A reference implementation is publicly available at https://github.com/davidkny22/dcaf-public.

1 Introduction

Neural networks implement behaviors through identifiable computational circuits—sparse subgraphs of model components that mediate specific input-output relationships. The field of mechanistic interpretability has made remarkable progress in discovering these circuits, from the manual reverse-engineering of the Indirect Object Identification (IOI) circuit in GPT-2 Small [Wang et al., 2023]—26 attention heads organized into 7 functional classes—to automated methods such as ACDC [Conmy et al., 2023], Attribution Patching [Syed et al., 2024], and Anthropic's Circuit Tracing [Ameisen et al., 2025]. The foundational vocabulary of residual streams, virtual weights, and QK/OV projection-level decomposition was established by Elhage et al. [2021], and the broader circuits hypothesis articulated by Olah et al. [2020].

Despite this progress, several limitations constrain current circuit discovery methodology. First, single-method circuit discovery exhibits high estimator variance: Méloux et al. [2025] demonstrate that EAP-IG circuit attributions have substantial variance under bootstrap resampling, undermining replicability. Second, the field has identified "interpretability illusions" arising from single-method conclusions and has explicitly called for cross-method validation [Sharkey et al., 2025].

Third, existing automated methods operate above the projection level—ACDC at head/edge granularity, CD-T at head-plus-position granularity [Hsu et al., 2025]—despite Elhage et al.'s (2021) foundational framework already defining projection-level structure. Fourth, activation patching results vary dramatically with hyperparameter choices [Zhang and Nanda, 2024], revealing methodological fragility. Fifth, no published framework uses paired training perturbations as primary discovery probes, despite growing evidence that training dynamics reveal circuit structure: Prakash et al. [2024] show that fine-tuning enhances existing mechanisms, and Tigges et al. [2024] demonstrate that circuit analyses are consistent across training and scale.

We introduce the Differential Circuit Analysis Framework (DCAF), which addresses these limitations through an integrative approach. DCAF synthesizes projection-level analysis [Elhage et al., 2021], training-as-probe methodology [Prakash et al., 2024, Tigges et al., 2024], the triangulation principle advocated by Long [2025] and Sharkey et al. [2025], and multi-method validation strategies emerging from the BlackboxNLP 2025 shared task [Mondorf et al., 2025] into a unified, formally specified protocol. Rather than introducing a single new technique, DCAF provides a complete pipeline that integrates independently established threads into a coherent framework with novel operational instruments: the 11-paired-objective probe configuration, the Direction Convergence Score (DCS), the 7-phase ablation protocol with interaction discovery, and adaptive tiered functional classification. These specific instruments have no direct named precedents in the literature.

The key contributions of this work are:

1. **Multi-signal training perturbation protocol.** We systematize the insights of Prakash et al. [2024] and Tigges et al. [2024] into a complete 11-signal configuration that uses preference optimization, supervised fine-tuning, gradient ascent, and sequential training as orthogonal discovery probes, extending contrastive methods [Panickssery et al., 2024, Arditi et al., 2024] from inference-time prompt contrasts to training-time signal contrasts.
2. **Triangulated confidence scoring.** We implement the triangulation principle called for by Long [2025] and Sharkey et al. [2025] through a specific operationalization across weight, activation, and geometric domains, producing calibrated confidence scores that reflect convergent evidence.
3. **Projection-level discovery at scale.** We operationalize at scale the QK/OV decomposition introduced by Elhage et al. [2021], extending it to gated MLP architectures and integrating it into an automated discovery pipeline alongside parameter-space methods such as APD [Braun et al., 2025].
4. **Ablation protocol with interaction discovery.** We specify a 7-phase ablation methodology that efficiently discovers individual, pairwise, and triple component interactions through seven parallel search strategies, producing functional classifications and interaction-enriched circuit graphs.
5. **Architecture-agnostic formulation.** The core methodology—perturbation experiments, multi-domain evidence, triangulated confidence—is defined independently of model architecture, with a complete transformer instantiation and preliminary sketches for vision and reinforcement learning systems.

A reference implementation of DCAF is publicly available at `https://github.com/davidkny22/dcaf-public`. The implementation covers the core mathematical framework, multi-path discovery, domain analysis modules (weight, activation, geometry), training diagnostics, confidence scoring, the 7-phase ablation protocol, circuit graph reconstruction, and CLI tooling for transformer architectures. Features currently under development include direction synthesis (global residual stream directions and DCS computation), MIB benchmark compatibility, and vision/diffusion model instantiation.

The remainder of this paper is organized as follows. Section 2 surveys related work. Sec-

tion 3 provides a framework overview. Sections 4–10 present the technical methodology in detail. Section 11 is reserved for empirical validation. Section 12 discusses limitations, and Section 13 concludes.

2 Related Work

DCAF builds on and extends a rich body of work in mechanistic interpretability, representation engineering, and AI safety. We organize prior art into seven areas and position DCAF relative to each.

2.1 Circuit Discovery Methods

Manual circuit analysis. Wang et al. [2023] reverse-engineered the IOI circuit in GPT-2 Small, identifying 26 attention heads in 7 functional classes through causal intervention. Hanna et al. [2023] similarly dissected the greater-than circuit, and Heimersheim and Janiak [2023] analyzed a Python docstring circuit in a 4-layer model. These studies demonstrated that transformer behaviors are implemented by identifiable circuits but required extensive manual effort.

Automated discovery. Conmy et al. [2023] introduced Automated Circuit DisCovery (ACDC), which scores edge importance via activation patching and prunes below a threshold, operating at head/MLP granularity. Syed et al. [2024] showed that Attribution Patching (EAP)—a linear approximation requiring only two forward passes and one backward pass—outperforms ACDC in edge recovery. Goldowsky-Dill et al. [2023] formalized path patching, the method ACDC automates. Hsu et al. [2025] achieved 97% ROC AUC with Contextual Decomposition for Transformers (CD-T), reducing circuit discovery runtime from hours to seconds. Li and Janson [2024] provided theoretical bounds for optimal ablation, producing smaller and more faithful circuits. Edin et al. [2025] introduced Gradient Interaction Modifications (GIM), a gradient-based method achieving high accuracy on the Mechanistic Interpretability Benchmark [Mueller et al., 2025]. Hadad et al. [2026] adapted neural network verification to circuit discovery with provable guarantees.

Production-scale methods. Anthropic's Circuit Tracing [Ameisen et al., 2025, Lindsey et al., 2025] introduced cross-layer transcoders and attribution graphs applied to Claude 3.5 Haiku. Lieberum et al. [2023] demonstrated circuit analysis at 70B scale.

DCAF complements these approaches by operating at the projection level within components (e.g., W_Q, W_K, W_V, W_O) rather than at the head/edge level, and by using multiple independent training perturbations as discovery probes rather than a single attribution method. DCAF's ablation protocol could integrate the theoretically grounded ablation techniques of Li and Janson [2024].

2.2 Representation Engineering and Steering Vectors

Zou et al. [2023] introduced Representation Engineering (RepE), examining population-level representations to monitor cognitive phenomena. Arditi et al. [2024] identified a single 1D subspace mediating refusal across 13 models up to 72B parameters. Li et al. [2023] used linear probes for Inference-Time Intervention on truthfulness. Marks and Tegmark [2024] established causal links between truth representations and outputs through the Geometry of Truth. Tigges et al. [2023] identified sentiment as a single linear dimension concentrating at syntactically neutral positions. Turner et al. [2023] proposed activation addition (ActAdd) for inference-time steering. Panickssery et al. [2024] systematized contrastive activation addition with paired positive/negative behavioral examples for Llama 2. Park et al. [2024] formalized the Linear Representation Hypothesis. Zhang

and Nanda [2024] demonstrated that activation patching results vary substantially with hyperparameter choices.

DCAF extends contrastive methods from inference-time prompt contrasts to training-time signal contrasts and requires convergence across multiple independent extraction methods. Where existing approaches extract directions from a single contrast pair, DCAF validates directions through cross-signal convergence: if multiple training signals independently produce directions that agree (as measured by DCS), the direction is robust to methodology choice.

2.3 Training Dynamics as an Interpretability Lens

This cluster represents the closest prior art to DCAF's training-perturbation methodology. Nanda et al. [2023] provided the earliest example of using training trajectories to expose circuit formation via progress measures for grokking. Prakash et al. [2024] introduced Cross-Model Activation Patching (CMAP) and showed that fine-tuning enhances existing mechanisms—the most direct precedent for training-as-discovery. Tigges et al. [2024] demonstrated that LLM circuit analyses are consistent across training and scale, providing evidence that training dynamics are a reliable lens. Wang et al. [2025] tracked edge changes across fine-tuning checkpoints. Singh et al. [2024] used optogenetics-inspired causal interventions throughout training to study induction head formation.

DCAF systematizes these insights into a complete multi-signal protocol, using 11 paired training objectives as orthogonal probes. Where prior work uses training dynamics as a post-hoc analytical lens, DCAF treats each training signal as a controlled perturbation experiment whose weight deltas, activation changes, and geometric effects are triangulated into confidence scores.

2.4 Weight-Space and Parameter-Level Analysis

Elhage et al. [2021] defined the QK and OV decomposition of attention heads—already operating at projection level—as the foundational analytical framework. Braun et al. [2025] introduced Attribution-based Parameter Decomposition (APD), the most explicit "weights-not-activations" circuit method, operating directly across all projection matrices. Gao et al. [2025] trained models with L_0-sparse weights to produce intrinsic projection-level circuits. Dooms et al. [2024] enabled weight-based mechanistic interpretability through bilinear MLPs. Golimblevskaia et al. [2025] introduced WeightLens and CircuitLens for interpreting features directly from learned weights.

DCAF does not claim to have invented projection-level analysis. Rather, it operationalizes at scale the QK/OV decomposition introduced by Elhage et al. [2021], extending it to gated MLP architectures (W_{gate}, W_{up}, W_{down}) and integrating it into an automated discovery pipeline. The framework's two-level hierarchy—projections as analysis units, components as graph nodes—allows sub-component weight analysis while maintaining compatibility with component-level activation and geometric methods.

2.5 Sparse Autoencoders and Feature Discovery

Cunningham et al. [2024] and Bricken et al. [2023] concurrently established sparse autoencoders (SAEs) as decomposition tools for language model activations. Gao et al. [2024] scaled SAEs to GPT-4 with 16M latents. Lieberum et al. [2024] released Gemma Scope, providing open-source SAEs across all layers of Gemma 2. Marks et al. [2025] extended SAEs to Sparse Feature Circuits, discovering causal subgraphs of interpretable features. Templeton et al. [2024] scaled monosemanticity to Claude 3 Sonnet. He et al. [2024] used dictionary learning as an alternative to activation patching for circuit discovery.

DCAF is complementary to SAE-based methods. SAEs decompose activations into interpretable features; DCAF identifies which components are behaviorally relevant through training perturbations. SAE features could serve as activation probes within DCAF's framework, and DCAF's weight-delta analysis operates orthogonally to SAE-based feature flow analysis.

2.6 Safety Circuit Protection

Zou et al. [2024] proposed Circuit Breakers for identifying and breaking representational pathways for undesirable behaviors. Han et al. [2025] identified safety neurons and proposed Safety Neuron Tuning, finding that edits to a sparse subset of safety neurons (often <5%, dropping to <1% for incremental safety dimensions) can reduce fine-tuning risks. Huang et al. [2025] introduced BOOSTER for defense against adversarial fine-tuning using safety-specific neuron detection. Zhu et al. [2025] developed SafetyLock, extracting safety-bias directions and re-injecting them post-fine-tuning. Wei et al. [2024] localized safety to ∼3% of parameters via pruning and low-rank modifications. Qi et al. [2024] established the foundational threat model showing that fine-tuning compromises safety. Lermen et al. [2023] demonstrated that LoRA fine-tuning efficiently undoes safety training.

DCAF provides the upstream discovery methodology that these protection methods require: identifying which projection matrices implement safety behaviors, at what confidence, and through which functional roles. The framework's opposition testing ($\mathcal{T}^+/\mathcal{T}^-$) directly reveals bidirectional control points—components where the same weights mediate both safe and unsafe behavior.

2.7 Methodological Foundations

Long [2025] formalized triangulation as an acceptance rule for mechanistic interpretability, explicitly advocating for multi-method validation. Méloux et al. [2025] reframed circuit discovery as a high-variance statistical estimation problem, recommending bootstrap-based uncertainty quantification. Mondorf et al. [2025] explored ensemble strategies for circuit localization at BlackboxNLP 2025, finding that combining methods improves performance. Geiger et al. [2021] developed causal abstractions of neural networks, and Geiger et al. [2025] unified activation patching, path patching, causal scrubbing, and distributed alignment search under a single theoretical framework. Belinkov [2022] provided a comprehensive analysis of probing classifiers, and Hewitt and Manning [2019] introduced the structural probe for syntax. Meng et al. [2022] localized factual recall to mid-layer MLPs. Rafailov et al. [2023] introduced Direct Preference Optimization, one of the training objectives DCAF uses as a discovery signal.

DCAF implements the triangulation principle through a specific operationalization: weight deltas capture what optimization modified, activation changes capture representational reorganization, and latent geometry captures behavioral direction structure. The DCS metric provides a concrete measure of the cross-signal convergence that Sharkey et al. [2025] call for and Méloux et al. [2025] demonstrate is needed.

3 Framework Overview

This section provides a high-level overview of the DCAF pipeline. Full technical details for each stage appear in Sections 4–10.

3.1 Operating Principle

DCAF treats each training signal as a controlled perturbation experiment. By analyzing what changes—in weights, activations, and latent geometry—when a model is trained under different behavioral pressures, it *discovers* candidate components, *maps* their causal connections, and *characterizes* their functional roles. Multiple independent signals and measurement domains provide both comprehensive characterization and confidence through convergence.

Definition 3.1 (Configurable Baseline Model)**.** *Let $\mathcal{M}_0$ be the baseline model with parameters $W_0 \in \mathbb{R}^d$ and components $\mathcal{K} = \{k_1, \ldots, k_m\}$. The baseline is chosen from two options based on the behavioral domain under study:*

Option A: *$\mathcal{M}_0 = \mathcal{M}_{pre}$ (the pretrained model directly). Appropriate when the target behavior's training data does not carry instruction-following signal.*

Option B (recommended for safety research): *$\mathcal{M}_0 = IT(\mathcal{M}_{pre}, \mathcal{D}_{IT})$—the pretrained model after instruction tuning on a behaviorally neutral dataset $\mathcal{D}_{IT}$. Recommended for any behavioral domain whose datasets inherently depend on instruction-following format.*

All weight deltas ΔW, activation baselines A_0, and ablation baseline restorations are relative to whichever $\mathcal{M}_0$ is selected.

Remark 3.1 (Rationale for Option B)**.** *Safety datasets teach two things simultaneously: (1) how to follow instructions and (2) the target safety behavior. If $\mathcal{M}_0$ is the pretrained model, every ΔW conflates "learned to follow instructions" with "learned the target behavior," causing DCAF to misidentify instruction-following circuits as safety circuits. Instruction-tuning first on a behaviorally neutral dataset means $\mathcal{M}_0$ already knows how to follow instructions, so behavioral training produces deltas isolating only the behavioral component.*

Definition 3.2 (Granularity Architecture)**.** *DCAF employs a two-level hierarchy:*

Analysis unit—Projection: *A single weight matrix within a model component. For attention components, the projections are W_Q, W_K, W_V, W_O (per head). For MLP components, the projections are $W_{gate}, W_{up}, W_{down}$ (per layer). Let $\mathcal{P}$ denote the set of all projections.*

Graph node—Component: *The functional unit for circuit graphs, ablation, and cross-domain confidence. Attention components are individual heads; MLP components are full MLP layers. These coincide with $\mathcal{K} = \{k_1, \ldots, k_m\}$.*

Projection-to-Component mapping: *$\mu_{proj} : \mathcal{P} \to \mathcal{K}$ maps each projection to its containing component.*

Remark 3.2 (Why Two Levels)**.** *Sub-component granularity is necessary because projections within the same component may show completely different behavioral signatures—W_Q and W_V within the same head may respond to entirely different training signals. Activation and geometry domains operate at component level because activations flow through the residual stream at head/MLP output granularity. This operationalizes at scale the QK/OV decomposition of Elhage et al. [2021], extending it to gated MLP architectures.*

3.2 Training Signals

Definition 3.3 (Training Signal Structure)**.** *Let $\mathcal{T} = \{t_1, \ldots, t_N\}$ be training transformations partitioned into:*

$$\mathcal{T}^+ = \{t : t \text{ reinforces target behavior}\} \tag{1}$$

$$\mathcal{T}^- = \{t : t \text{ reinforces opposite behavior}\} \tag{2}$$

$$\mathcal{T}^0 = \{t : t \text{ is behaviorally neutral baseline}\} \tag{3}$$

The $\mathcal{T}^+/\mathcal{T}^-$ structure extends contrastive methods [Panickssery et al., 2024, Arditi et al., 2024] from inference-time prompt contrasts to training-time signal contrasts, testing whether components show opposing responses to opposing training pressures.

Definition 3.4 (Canonical 11-Signal Instantiation)**.** *The 11 canonical signals, where $a \succ b$ denotes preference (chosen = a, rejected = b):*

Target Behavior Cluster $\mathcal{T}^+$:

$$
\begin{aligned}
t_1 &= \mathit{PrefOpt}(\mathit{target} \succ \mathit{opposite}) && \textit{(preference optimization, e.g., SimPO, DPO)}\\
t_2 &= \mathit{SFT}(\mathit{target}) && \textit{(supervised fine-tuning)}\\
t_3 &= \mathit{Cumulative}(\mathit{target}) && \textit{(SFT then PrefOpt sequentially)}\\
t_4 &= \mathit{Anti}(\mathit{opposite} \succ \mathit{target}) && \textit{(gradient ascent on preference margin from base)}\\
t_5 &= \mathit{Negated}^{(-)}(\mathit{opposite} \succ \mathit{target}) && \textit{(gradient ascent from opposite-checkpoint)}
\end{aligned}
$$

Opposite Behavior Cluster $\mathcal{T}^-$:

$$
\begin{aligned}
&t_6 = \mathit{PrefOpt}(\mathit{opposite} \succ \mathit{target}), \quad t_7 = \mathit{SFT}(\mathit{opposite}), \quad t_8 = \mathit{Cumulative}(\mathit{opposite})\\
&t_9 = \mathit{Anti}(\mathit{target} \succ \mathit{opposite}), \quad t_{10} = \mathit{Negated}^{(+)}(\mathit{target} \succ \mathit{opposite})
\end{aligned}
$$

Neutral Baseline $\mathcal{T}^0$: $t_{11} = \mathit{DomainNative}(\mathit{neutral})$

Remark 3.3 (Minimal Signal Set)**.** *The full 11-signal protocol maximizes discovery coverage and opposition measurement rigor. For computational efficiency, a minimal 3-signal subset—t_1 (PrefOpt target), t_6 (PrefOpt opposite), t_{11} (neutral)—captures the core $\mathcal{T}^+/\mathcal{T}^-/\mathcal{T}^0$ structure. Expand to the full set when computational budget allows.*

3.3 Pipeline Summary

The complete DCAF pipeline proceeds as follows:

$$
\mathcal{M}_0 \xrightarrow{\mathcal{T}} \left\{ \begin{array}{c} \Delta W \to \mathcal{H}_W \\ \Delta A \to \mathcal{H}_A \\ \nabla \mathcal{L} \to \mathcal{H}_G \end{array} \right\} \xrightarrow{\cup} \mathcal{H}_{\text{disc}} \xrightarrow{C_W, C_A, C_G} \mathcal{H}_{\text{cand}} \xrightarrow{\text{ablation}} \mathcal{H}_{\text{conf}} \to G, \text{class}(k) \tag{4}
$$

Concretely:

0. **Baseline selection:** Choose $\mathcal{M}_0$ (Section 3).
1. **Signal training:** Run all training signals from $\mathcal{M}_0$ with periodic checkpoint evaluation.
2. **Peak checkpoint selection:** For each signal, identify the confirmed peak checkpoint $W_{i,\text{peak}}$ using stability-confirmed maximum behavioral metrics (Section 4).
3. **Weight delta computation:** Compute $\Delta W_i = W_{i,\text{peak}} - W_0$ for all signals.
4. **Signal effectiveness:** Compute $\text{eff}(i)$ from baseline to peak-checkpoint metrics.
5. **Multi-path discovery:** Weight $\to \mathcal{H}_W$, Activation $\to \mathcal{H}_A$, Gradient $\to \mathcal{H}_G$ (Section 4).
6. **Domain analysis:** Weight, activation, and geometry analysis (Sections 5–5.3).
7. **Training diagnostics:** Activation delta alignment, optional curvature tracking (Section 6).
8. **Unified confidence:** Triangulated scoring with multi-path bonus (Section 7).
9. **Circuit graph reconstruction:** Edge discovery via 5 methods (Section 8).
10. **Steering vector optimization:** Bidirectional steering and validation (Section 9).
11. **Ablation and functional classification:** 7-phase protocol (Section 10).

Symbol	Meaning
$\mathcal{M}_0$	Configured baseline model
$\mathcal{K}$, k	Set of components; single component
$\mathcal{P}$, proj	Set of projections; single projection matrix
$\mu_{\text{proj}} : \mathcal{P} \to \mathcal{K}$	Projection-to-component mapping
$\mathcal{T}^+, \mathcal{T}^-, \mathcal{T}^0$	Target, opposite, and neutral signal clusters
$\Delta W_i^{(\text{proj})}$	Weight delta: $W_{i,\text{peak}}^{(\text{proj})} - W_0^{(\text{proj})}$
$\text{eff}(i)$	Signal effectiveness score $\in [0, 1]$
$\mathcal{H}_W, \mathcal{H}_A, \mathcal{H}_G$	Weight-, activation-, gradient-discovered sets
$\mathcal{H}_{\text{disc}}, \mathcal{H}_{\text{cand}}, \mathcal{H}_{\text{conf}}$	Discovered, candidate, confirmed component sets
$C_W^{(k)}, C_A^{(k)}, C_G^{(k)}$	Weight, activation, geometric confidence
$C^{(k)}$	Unified confidence
$\text{DCS}(L)$	Direction Convergence Score at layer L
$G = (V, E)$	Behavioral circuit graph

Table 1: Key notation. Full reference in Appendix C.

Notation. Table 1 provides a compact reference for the most important symbols. A full notation reference appears in Appendix C.

4 Multi-Path Discovery

DCAF employs three complementary discovery paths. Each path identifies candidate projections using different evidence, which are then mapped to components via μ_{proj}. All discovered components then receive confidence scores from all three measurement domains.

4.1 Peak Checkpoint Selection

Training runs do not always improve monotonically. A signal may achieve its strongest behavioral effect mid-training before degrading.

Definition 4.1 (Peak Checkpoint)**.** *For training signal t_i, let $\{W_{i,\tau}\}_{\tau=1}^{T}$ be the sequence of checkpoints. The peak checkpoint $W_{i,peak}$ is the checkpoint at which the signal-specific behavioral metric $m_{i,\tau}$ achieves its confirmed maximum:*

$$W_{i,peak} = W_{i,\tau^*}, \qquad \tau^* = \arg\max_{\tau} \; m_{i,\tau} \quad \textit{subject to Confirmed}(\tau) \tag{5}$$

where confirmation prevents selecting transient spikes:

$$\textit{Confirmed}(\tau) = \left(m_{i,\tau} = \max_{\tau' \leq \tau} m_{i,\tau'}\right) \;\wedge\; \left(\frac{1}{K}\sum_{j=1}^{K} m_{i,\tau+j} \geq m_{i,\tau} - \delta \cdot m_{i,\tau}\right) \tag{6}$$

with confirmation window $K = 3$ and stability tolerance $\delta = 0.05$.

Definition 4.2 (Weight Delta)**.**

$$\Delta W_i^{(proj)} = W_{i,peak}^{(proj)} - W_0^{(proj)} \tag{7}$$

Each delta is a full matrix of the same shape as the projection weight matrix.

4.2 Significance Predicates

Definition 4.3 (Percentile Threshold).

$$\Phi_k(\{\|\Delta W_i^{(proj)}\|_{RMS}\}_{proj\in\mathcal{P}}) = \inf\left\{\theta : \frac{|\{proj : \|\Delta W_i^{(proj)}\|_{RMS} \le \theta\}|}{|\mathcal{P}|} \ge \frac{k}{100}\right\} \tag{8}$$

Definition 4.4 (Significance Predicates).

$$sig(proj, i) = \mathbf{1}\left[\|\Delta W_i^{(proj)}\|_{RMS} \ge \Phi_{\tau_{sig}}(\cdot)\right] \tag{9}$$

$$\overline{sig}(proj, i) = \mathbf{1}\left[\|\Delta W_i^{(proj)}\|_{RMS} < \Phi_{\tau_{base}}(\cdot)\right] \tag{10}$$

with $\tau_{sig} = 85$ (top 15% of delta norms) and $\tau_{base} = 50$ (median).

4.3 Path 1: Weight-Based Discovery ($\mathcal{H}_W$)

Weight-based discovery identifies projections that changed substantially during behavioral training—the most direct evidence of optimization pressure.

Definition 4.5 (Signal Effectiveness).

$$eff(i) = normalize_{[0,1]}\left(\Delta_{improve}(i) + \beta \cdot threshold(i)\right) \tag{11}$$

where $\Delta_{improve}$ is the normalized improvement metric (loss reduction for SFT, margin improvement for preference methods, reward improvement for RL), threshold(i) is a binary bonus for crossing a behavioral threshold, and $\beta = 0.4$.

Definition 4.6 (Aggregated Cluster Deltas).

$$\bar{\delta}_+^{(proj)} = \frac{\sum_{i\in\mathcal{T}^+} eff(i) \cdot \Delta W_i^{(proj)}}{\sum_{i\in\mathcal{T}^+} eff(i)}, \quad \bar{\delta}_-^{(proj)} = \frac{\sum_{j\in\mathcal{T}^-} eff(j) \cdot \Delta W_j^{(proj)}}{\sum_{j\in\mathcal{T}^-} eff(j)} \tag{12}$$

Definition 4.7 (Opposition Degree).

$$cos_opp^{(proj)} = \cos\Big(flatten(\bar{\delta}_+^{(proj)}),\ flatten(\bar{\delta}_-^{(proj)})\Big) \tag{13}$$

$$opp_degree(proj) = \max\Big(0,\ -cos_opp^{(proj)}\Big) \tag{14}$$

When $\mathcal{T}^+$ and $\mathcal{T}^-$ push the projection in opposite directions, opp_degree ≈ 1. Cosine similarity is self-normalizing, requiring no layer-specific calibration.

Definition 4.8 (Weight-Based Discovery Score).

$$S_W^{(proj)} = \min\left(1,\ \frac{\sum_{i\in\mathcal{T}\setminus\mathcal{T}^0} eff(i)^q \cdot sig(proj, i)}{\sum_{i\in\mathcal{T}\setminus\mathcal{T}^0} eff(i)^q} + \alpha \cdot opp_degree(proj)\right) \cdot \mathbf{1}\left[\forall t \in \mathcal{T}^0 : \overline{sig}(proj, t)\right] \tag{15}$$

with $q = 2$ (effectiveness power) and $\alpha = 0.2$ (opposition bonus weight). The baseline filter excludes projections responding to neutral training.

Definition 4.9 (Weight-Based Discovery Set).

$$\mathcal{H}_W = \{proj \in \mathcal{P} : S_W^{(proj)} \ge \tau_W\} \tag{16}$$

where τ_W is the weight discovery threshold (default: 0.3).

4.4 Path 2: Activation-Based Discovery ($\mathcal{H}_A$)

Activation-based discovery identifies **leverage points**: projections where small weight changes produce large representational effects, missed by weight-based discovery.

Definition 4.10 (Component Activation Magnitude)**.**

$$m_{agg}^{(k)} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \frac{1}{|\Pi|} \sum_{\pi \in \Pi} m_i^{(k,\pi)} \tag{17}$$

where $\Pi = \{\pi_{detect}, \pi_{decide}, \pi_{eval}\}$ *are probe types capturing behavioral processing at different stages.*

Definition 4.11 (Two-Stage Discovery)**.** *Flag components with notable activation changes using a generous threshold ($\tau_{comp} = 70$th percentile), then filter projections within flagged components using a strict threshold ($\tau_{act} = 85$th percentile):*

$$\mathcal{H}_A = \left\{ proj \in \mathcal{P} : \mu_{proj}(proj) \in \mathcal{K}_{flagged} \wedge S_A^{(proj)} \geq \Phi_{\tau_{act}}(\cdot) \right\} \tag{18}$$

4.5 Path 3: Gradient-Based Discovery ($\mathcal{H}_G$)

Gradient-based discovery identifies projections with high behavioral gradients: projections that *would* affect behavior if modified, regardless of whether training actually changed them.

Definition 4.12 (Behavioral Gradient)**.**

$$g^{(proj)} = \sum_{i \in \mathcal{T}^+ \cup \mathcal{T}^-} eff(i) \cdot \left\| \frac{\partial \mathcal{O}_i}{\partial W^{(proj)}} \right\|_F \tag{19}$$

Definition 4.13 (Gradient-Based Discovery Set)**.**

$$\mathcal{H}_G = \left\{ proj \in \mathcal{P} : g^{(proj)} \geq \Phi_{\tau_{grad}}(\cdot) \right\} \tag{20}$$

with $\tau_{grad} = 85$.

4.6 Discovery Integration

Definition 4.14 (Unified Discovery Set)**.**

$$\mathcal{H}_{disc} = components(\mathcal{H}_W) \cup components(\mathcal{H}_A) \cup components(\mathcal{H}_G) \tag{21}$$

Discovery is inclusive: a component flagged by any *path becomes a candidate for confidence scoring and ablation validation. The three paths are deliberately independent, so multi-path discovery (paths(k) ≥ 2) indicates converging evidence from independent methods.*

5 Domain Analysis

All discovered components ($\mathcal{H}_{\text{disc}}$) receive confidence scores from three measurement domains, regardless of which path discovered them.

5.1 Weight Analysis

Weight analysis operates at projection level. Full methodology details including RMS normalization, SVD diagnostics, and signal effectiveness formulas appear in the definitions above and in Appendix A.

Definition 5.1 (RMS-Normalized Frobenius Norm)**.**

$$\|\Delta W_i^{(proj)}\|_{RMS} = \frac{\|\Delta W_i^{(proj)}\|_F}{\sqrt{m \cdot n}} \tag{22}$$

where (m, n) is the projection matrix shape. Dividing by $\sqrt{m \cdot n}$ eliminates dimensional bias—an MLP gate projection (2304×9216) would otherwise produce $\sim$6$\times$ larger norms than an attention Q projection (2304×256) for identical per-element changes.

Definition 5.2 (Weight Confidence C_W)**.**

$$C_W^{(proj)} = \min\left(1, \frac{\sum_{i \in \mathcal{T} \setminus \mathcal{T}^0} \mathit{eff}(i)^q \cdot \mathit{sig}(proj, i)}{\sum_{i \in \mathcal{T} \setminus \mathcal{T}^0} \mathit{eff}(i)^q} + \alpha \cdot \mathit{opp_degree}(proj)\right) \cdot \mathbf{1}\left[\forall t \in \mathcal{T}^0 : \overline{\mathit{sig}}(proj, t)\right] \tag{23}$$

Component-level: $C_W^{(k)} = \max_{proj \in \mu_{proj}^{-1}(k)} C_W^{(proj)}$.

5.1.1 SVD Diagnostics on Projection Deltas

Definition 5.3 (SVD Diagnostics)**.** *Compute the SVD of aggregated cluster delta matrices. Two derived metrics characterize the structure of weight changes:*

Rank-1 fraction: *rank1(proj) $= \sigma_1^2 / \sum_i \sigma_i^2$—how concentrated the weight change is along a single input-to-output mapping.*

Spectral opposition: *spec_opp(proj) $= -\cos(V_{h,+}[0], V_{h,-}[0])$—whether $\mathcal{T}^+$ and $\mathcal{T}^-$ oppose along the primary axis of change.*

These diagnostics are optional and do not enter the confidence formula; they provide interpretive depth for projections already flagged as significant.

5.2 Activation Analysis

Definition 5.4 (Probe Types)**.** *Three probe types capture behavioral processing at different stages:*

$\pi_{\mathbf{detect}}$**:** ***Behavioral trigger detection*** *(upstream)—does the model encode/recognize behavioral content?*

$\pi_{\mathbf{decide}}$**:** ***Behavioral decision point*** *(middle)—does the model make the behavioral choice?*

$\pi_{\mathbf{eval}}$**:** ***Behavioral path evaluation*** *(parallel)—does the model prefer the behavioral path?*

Definition 5.5 (Activation Confidence C_A)**.**

$$n_A^{(k)} = \sum_{i \in \mathcal{T}} \sum_{\pi \in \Pi} \mathit{sig}_A(k, i, \pi), \qquad C_A^{(k)} = \frac{n_A^{(k)}}{|\mathcal{T}| \cdot |\Pi|} \tag{24}$$

where $\mathit{sig}_A(k, i, \pi) = \mathbf{1}[m_i^{(k,\pi)} \geq \Phi_{\tau_{act}}^A(\cdot)]$ and $m_i^{(k,\pi)}$ is the average ℓ_2 norm of activation change across probe examples.

5.3 Geometry Analysis

The geometry domain analyzes how behavior is represented in activation space, measuring whether training created or strengthened directions that separate behavioral classes.

5.3.1 Contrastive Direction Extraction

Definition 5.6 (Whitened Contrastive Direction)**.**

$$\mu^+ = \mathbb{E}[A^+], \quad \mu^- = \mathbb{E}[A^-], \quad \Sigma_w = \frac{1}{2}\left(Cov(A^+) + Cov(A^-)\right) \tag{25}$$

$$d_i^{(k)} = \frac{(\Sigma_w + \lambda I)^{-1}(\mu^+ - \mu^-)}{\|(\Sigma_w + \lambda I)^{-1}(\mu^+ - \mu^-)\|} \tag{26}$$

5.3.2 Cross-Signal Alignment

Definition 5.7 (LRS Components)**.**

$$coh_+^{(k)} = \frac{1}{|\mathcal{T}^+|^2} \sum_{i,j \in \mathcal{T}^+} \cos(d_i^{(k)}, d_j^{(k)}) \qquad \textit{(target cluster coherence)} \tag{27}$$

$$coh_-^{(k)} = \frac{1}{|\mathcal{T}^-|^2} \sum_{i,j \in \mathcal{T}^-} \cos(d_i^{(k)}, d_j^{(k)}) \qquad \textit{(opposite cluster coherence)} \tag{28}$$

$$opp^{(k)} = -\frac{1}{|\mathcal{T}^+||\mathcal{T}^-|} \sum_{i \in \mathcal{T}^+, j \in \mathcal{T}^-} \cos(d_i^{(k)}, d_j^{(k)}) \qquad \textit{(cross-cluster opposition)} \tag{29}$$

$$orth^{(k)} = 1 - \frac{1}{|\mathcal{T}^+ \cup \mathcal{T}^-|} \sum_i \left|\cos(d_i^{(k)}, d_{t_0}^{(k)})\right| \qquad \textit{(baseline orthogonality)} \tag{30}$$

Definition 5.8 (Linear Representation Score (LRS))**.** *A Hölder (power) mean of six geometric metrics:*

$$LRS^{(k)} = \left(\frac{\sum_{i=1}^{6} w_i \cdot (x_i^{(k)} + \varepsilon)^p}{\sum_{i=1}^{6} w_i}\right)^{1/p} \tag{31}$$

with $p = 0.5$, $\varepsilon = 0.05$, uniform weights. The six components are: coh_+, coh_-, opp, orth, confound independence ρ_c, and scaled predictivity gain.

Definition 5.9 (Geometric Confidence C_G)**.**

$$C_G^{(k)} = LRS^{(k)} \cdot gen^{(k)} \tag{32}$$

where $gen^{(k)}$ is the ratio of out-of-distribution to within-distribution predictivity.

5.3.3 Direction Convergence Score (DCS)

DCS measures how strongly the training signals agree on a behavioral direction at each layer of the residual stream.

Definition 5.10 (DCS Convergence Components)**.** *For layer L, compute three convergence metrics from per-signal contrastive directions extracted from residual stream activations:*

$$coh_+^{(L)} = \frac{1}{\binom{|\mathcal{T}^+|}{2}} \sum_{\substack{i,j \in \mathcal{T}^+ \\ i<j}} |\cos(d_i^{(L)}, d_j^{(L)})| \tag{33}$$

$$coh_-^{(L)} = \frac{1}{\binom{|\mathcal{T}^-|}{2}} \sum_{\substack{i,j \in \mathcal{T}^- \\ i<j}} |\cos(d_i^{(L)}, d_j^{(L)})| \tag{34}$$

$$opp^{(L)} = \frac{1}{|\mathcal{T}^+||\mathcal{T}^-|} \sum_{\substack{i \in \mathcal{T}^+ \\ j \in \mathcal{T}^-}} \left(-\cos(d_i^{(L)}, d_j^{(L)})\right) \tag{35}$$

Definition 5.11 (DCS Aggregation)**.**

$$DCS(L) = \left[\frac{1}{6} \sum_{x \in \{coh_+, coh_-, opp, ortho, confound, pred\}} (x^{(L)} + \varepsilon)^p\right]^{1/p} \tag{36}$$

with $p = 0.5$, $\varepsilon = 0.05$, where the three validation components (ortho, confound, pred) are computed only when the convergence prerequisite is met (all three convergence metrics ≥ 0.3).

Definition 5.12 (Layer Trajectory)**.**

$$L_{emerge} = \min\{L : DCS(L) > \tau_{dcs\text{-}peak}\} \qquad \textit{(emergence layer)} \tag{37}$$

$$L^* = \arg\max_L DCS(L) \qquad \textit{(peak layer)} \tag{38}$$

$$stab(L) = \cos(d_{consensus}^{(L)}, d_{consensus}^{(L+1)}) \qquad \textit{(direction stability)} \tag{39}$$

5.3.4 Nonlinear Diagnostics

When $\text{LRS}^{(k)} < 0.4$, the behavioral representation may be nonlinear. Rather than discarding the component, DCAF triggers diagnostic methods: PaCMAP embedding with silhouette scoring, Procrustes alignment for structural mirror scoring, degree-2 polynomial probing, and kernel LDA. These diagnostics are purely informational—they characterize the nonlinearity without altering confidence scores. Full details appear in Appendix C.

6 Training Diagnostics

Training diagnostics measure the quality of the training process itself. Where domain analyses answer "which projections implement the behavior?", training diagnostics answer "did the training signals behave as expected?"

Definition 6.1 (Activation Delta Alignment)**.** *For component k, three alignment metrics from the*

activation deltas $\Delta A_i^{(k)} = A_{i,peak}^{(k)} - A_{i,pre}^{(k)}$*:*

$$align_+^{(k)} = \frac{1}{|\mathcal{T}^+|^2} \sum_{i,j \in \mathcal{T}^+} \cos(\Delta A_i^{(k)}, \Delta A_j^{(k)}) \tag{40}$$

$$align_-^{(k)} = \frac{1}{|\mathcal{T}^-|^2} \sum_{i,j \in \mathcal{T}^-} \cos(\Delta A_i^{(k)}, \Delta A_j^{(k)}) \tag{41}$$

$$oppose_\Delta^{(k)} = \frac{1}{|\mathcal{T}^+||\mathcal{T}^-|} \sum_{i \in \mathcal{T}^+, j \in \mathcal{T}^-} \cos(\Delta A_i^{(k)}, \Delta A_j^{(k)}) \tag{42}$$

Expected values: $align_+ > 0.7$*,* $align_- > 0.7$*,* $oppose_\Delta < -0.5$*.*

Definition 6.2 (Online Curvature (Optional))**.** *For training signal* t_i *at component* k*, track the activation trajectory curvature:*

$$\kappa_i^{(k)} = \frac{path_length_i^{(k)}}{direct_distance_i^{(k)}} - 1 \tag{43}$$

$\kappa = 0$*: training moved representations directly toward target. High* κ*: competing gradients or unstable optimization. Asymmetric* κ *(*$\mathcal{T}^+$ *low,* $\mathcal{T}^-$ *high): one behavioral direction is easier to learn at this component.*

7 Confidence and Candidate Selection

Definition 7.1 (Base Triangulated Confidence)**.** *For component* $k \in \mathcal{H}_{disc}$*:*

$$C_{base}^{(k)} = \left[(C_W^{(k)} + \varepsilon)^w \cdot (C_A^{(k)} + \varepsilon) \cdot (C_G^{(k)} + \varepsilon) \right]^{\frac{1}{w+2}} \tag{44}$$

with domain weight $w = 2$ *and smoothing* $\varepsilon = 0.05$*. The weighted geometric mean ensures all domains contribute, domain agreement amplifies confidence, and the formula remains bounded in* $[0, 1]$*.*

Definition 7.2 (Multi-Path Discovery Bonus)**.**

$$bonus(k) = \beta_{path} \cdot \max(0, paths(k) - 1) \tag{45}$$

with $\beta_{path} = 0.15$*. Additive rather than multiplicative to avoid penalizing single-path discoveries.*

Definition 7.3 (Final Unified Confidence)**.**

$$C^{(k)} = \min(1, C_{base}^{(k)} + bonus(k)) \tag{46}$$

Definition 7.4 (Candidate and Confirmed Sets)**.**

$$\mathcal{H}_{cand} = \{k \in \mathcal{H}_{disc} : C^{(k)} \geq \tau_{unified}\} \tag{47}$$

$$\mathcal{H}_{conf} = \{k \in \mathcal{H}_{cand} : ablation\ confirms\ behavioral\ relevance\} \tag{48}$$

Remark 7.1 (Domain Disagreement Diagnostics)**.** *When domains disagree, the pattern reveals the component's nature: weight high / activation low indicates weights changed without representational reorganization; weight low / activation high indicates a leverage point; geometry high / others low indicates a pre-existing direction that training leveraged.*

8 Circuit Graph Reconstruction

Definition 8.1 (Circuit Nodes)**.** $V = \mathcal{H}_{conf}$

Definition 8.2 (Edge Discovery Methods)**.** *Five complementary methods:*

1. ***Activation flow:*** $corr^{(i)}(k, k') = Corr(A_i^{(k)}, A_i^{(k')})$
2. ***Targeted ablation:*** $E^{abl}_{k\to k'} = \|A^{(k')}_{with\ k} - A^{(k')}_{without\ k}\|_F$
3. ***Targeted steering:*** $E^{steer}_{k\to k'} = \|A^{(k')}_{with\ v_k} - A^{(k')}_{without\ v_k}\|_F$
4. ***Gradient flow:*** $grad_corr(k, k') = Corr(\nabla_{W_k}\mathcal{L}, \nabla_{W_{k'}}\mathcal{L})$
5. ***Cross-layer attention:*** $attn_flow(k_\ell, k_{\ell'}) = \sum_s P_{\ell',h}[s, pos(k_\ell)]$

Definition 8.3 (Causal Edge and Circuit Graph)**.**

$$E_{k\to k'} = \max\left(E^{abl}_{k\to k'}, E^{steer}_{k\to k'}\right), \qquad G = \left(V, \{(k, k') : E_{k\to k'} \geq \tau_E\}\right) \tag{49}$$

Definition 8.4 (Edge Pathway Attribution)**.** *For edges targeting an attention head, determine which input pathway carries the causal signal by measuring* $\|\Delta A_Q\|_F$*,* $\|\Delta A_K\|_F$*,* $\|\Delta A_V\|_F$ *under source ablation, normalized to weights summing to 1.0.*

9 Steering Vector Optimization

Steering vectors validate the causal role of discovered components by directly intervening in the model's computations.

Definition 9.1 (Bidirectional Steering Vectors)**.** *For component* k*, optimize target and opposite steering vectors:*

$$v^*_{k,+} = \arg\max_v \mathcal{L}_{target}(\mathcal{M}_0 + v \cdot e_k) \tag{50}$$

$$v^*_{k,-} = \arg\max_v \mathcal{L}_{opposite}(\mathcal{M}_0 + v \cdot e_k) \tag{51}$$

Definition 9.2 (Steering-Geometry Alignment)**.**

$$\alpha^{(k)}_+ = \cos(v^*_{k,+},\ d^{(k)}) \qquad \textit{(target steering vs. geometry)} \tag{52}$$

$$\alpha^{(k)}_- = \cos(v^*_{k,-},\ d^{(k)}) \qquad \textit{(opposite steering vs. geometry)} \tag{53}$$

$$\alpha^{(k)}_\pm = \cos(v^*_{k,+},\ v^*_{k,-}) \qquad \textit{(target vs. opposite steering)} \tag{54}$$

$\alpha_\pm \approx -1$*: single bidirectional control axis.* $\alpha_\pm \neq -1$*: the component mediates each behavioral direction through partially different mechanisms, enabling distinct reinforcement and pathway-blocking intervention strategies.*

Definition 9.3 (Steering Validation Protocol)**.** *Five steps applied to both* $v^*_{k,+}$ *and* $v^*_{k,-}$*: (1) behavioral test, (2) scaling ablation at magnitudes* $\{1.0, 0.7, 0.5, 0.3, 0.1, 0.0\}$ *for dose-response, (3) component ablation for necessity assessment, (4) downstream activation analysis, and (5) cross-reference with DCAF-identified circuit nodes.*

10 Ablation and Functional Classification

10.1 Ablation Methodology

Two ablation approaches are used: **mean ablation** (replace activations with mean across a reference set) and **baseline restoration** (replace with baseline model $\mathcal{M}_0$ values).

Definition 10.1 (Behavioral Relevance Classification)**.**

$$relevance(k) = \begin{cases} GENERAL & \text{if } func_{ablated} < 0.5 \cdot func_{intact} \\ BEHAVIORAL & \text{if } I_\pi^{(k)} > 0.1 \text{ for any } \pi \\ NONE & \text{otherwise} \end{cases} \tag{55}$$

where $I_\pi^{(k)} = |score_{intact}^\pi - score_{ablated}^\pi| / (|score_{intact}^\pi| + \varepsilon)$.

10.2 Seven-Phase Interaction Discovery Protocol

Starting from $\mathcal{H}_{\text{cand}}$, the protocol discovers component interactions through seven phases:

Phase 1: Individual ablation. Test every candidate individually (∼30–50 tests). Partition into $\mathcal{H}_{\text{solo}}$ (behavioral impact alone) and $\mathcal{H}_{\overline{\text{solo}}}$ (no individual impact).

Phase 2: Seven parallel pair/group strategies. Each strategy independently identifies pairs likely to interact, with total budget $B_{\text{pair}} = 300$:

A: Graph-adjacent pairs from preliminary circuit edges.
B: Gradient-based screening via $\text{interact}(k_1, k_2) = \|\nabla_{k_2}\mathcal{L}|_{k_1 \text{ ablated}} - \nabla_{k_2}\mathcal{L}|_{\text{intact}}\|$.
C: Activation correlation pairs from concatenated activation delta vectors.
D: Hierarchical clustering with average linkage on correlation matrix.
E: Opposition grouping of components with similar opposition profiles.
F: Cross-layer composition of attention heads from different layers.
G: Random sampling ($n = 200$), confidence-weighted, as false-negative safeguard.

Phase 3: Refinement. Targeted follow-up: solo×solo pairs, no-solo pair members with solo components, flagged Hessian pairs for triple testing.

Phase 4: Cross-validation counting. Track $\text{discovery_count}(k)$—how many strategies found each component in a significant pair. ≥ 2: high confidence. $= 1$: re-test.

Phase 5: Triple detection. Capped at $B_{\text{triple}} = 100$ tests. Two approaches: extend Hessian pairs with high-confidence third members; combine no-effect components with top solo components.

Phase 6: Orphan analysis. High-confidence orphans ($C^{(k)} > 0.6$) re-tested with solo components. Low-confidence orphans flagged for optional exhaustive testing.

Phase 7: Final classification. Assign interaction requirement (SOLO/PAIR/GATE/ORPHAN) and interaction type:

$$\text{SYNERGISTIC:} \quad I(k_1, k_2) > I(k_1) + I(k_2) + \varepsilon_{\text{syn}} \tag{56}$$

$$\text{ADDITIVE:} \quad |I(k_1, k_2) - (I(k_1) + I(k_2))| \leq \varepsilon_{\text{syn}} \tag{57}$$

$$\text{REDUNDANT:} \quad I(k_1, k_2) \approx \max(I(k_1), I(k_2)) \tag{58}$$

The confirmed set: $\mathcal{H}_{\text{conf}} = \{k \in \mathcal{H}_{\text{cand}} : \text{classification}(k) \neq \text{ORPHAN}\}$.

10.3 Adaptive Tiered Functional Classification

After ablation confirms behavioral relevance, components are classified by probe-type impact using an adaptive three-tier system.

Definition 10.2 (Adaptive Threshold)**.** *Let* $I^{(k)}_{\max} = \max\{I^{(k)}_{detect}, I^{(k)}_{decide}, I^{(k)}_{eval}\}$*. Sort impacts* $I_{(1)} \geq I_{(2)} \geq I_{(3)}$*. Define tight clustering:* $tight(k) = \mathbf{1}[\max\{g_1, g_2\} < \tau_{gap}]$ *where* $g_i = I_{(i)} - I_{(i+1)}$*.*

$$r_{primary}(k) = r^{relaxed}_{primary} \cdot tight(k) + r^{strict}_{primary} \cdot (1 - tight(k)) \tag{59}$$

with $r^{strict} = 0.8$*,* $r^{relaxed} = 0.7$*.*

Definition 10.3 (Tier Assignment)**.** *Primary:* $P(k) = \{\pi : I^{(k)}_{\pi} \geq r_{primary}(k) \cdot I^{(k)}_{\max} \wedge I^{(k)}_{\pi} \geq \tau_{abs}\}$*. Auxiliary:* $A(k) = \{\pi : I^{(k)}_{\pi} \geq r_{aux} \cdot I^{(k)}_{\max} \wedge \pi \notin P(k)\}$ *(only if* $P(k) \neq \emptyset$*).*

Functional roles correspond to probe types: ***Recognition*** *(*π_{detect}*),* ***Steering*** *(*π_{decide}*),* ***Preference*** *(*π_{eval}*).*

11 Experiments

12 Discussion and Limitations

DCAF provides a formally specified protocol integrating projection-level analysis, multi-signal training perturbations, and triangulated confidence scoring into a unified pipeline for neural circuit discovery. We discuss its current limitations honestly.

Computational cost. The 11-signal protocol requires 11 training runs, each producing checkpoints that must be evaluated for peak selection. This is substantially more expensive than single-method approaches such as ACDC (one forward/backward pass per edge) or EAP (two forward passes and one backward pass). We mitigate this through three mechanisms: the 3-signal minimal protocol (t_1, t_6, t_{11}) captures the core $\mathcal{T}^+/\mathcal{T}^-/\mathcal{T}^0$ structure at one-quarter the cost; signal training can piggy-back on existing fine-tuning pipelines; and baseline model activations ($\mathcal{M}_0$) are cached and shared across all signals. Nevertheless, the full protocol is best suited for high-stakes analyses where confidence in circuit identification justifies the computational investment.

Empirical validation. As of this writing, the framework has been implemented and tested on individual modules, but systematic end-to-end validation on production-scale models with comparison to established benchmarks (particularly MIB [Mueller et al., 2025]) is ongoing. We position this work as a formally specified methodological protocol in the spirit of pre-registration—making explicit the complete analysis procedure so that empirical instantiation can be evaluated against a fixed specification. This approach has precedent in mechanistic interpretability: Elhage et al. [2021] established the field's mathematical vocabulary primarily through theoretical exposition, and Sharkey et al. [2025] published an 89-page research agenda without new experiments.

Architecture-specific instantiations. The core framework (Part I) is architecture-agnostic, but the complete transformer instantiation is the only validated domain-specific implementation. Vision, diffusion, and reinforcement learning instantiations remain preliminary sketches. Extending the probe type definitions (π_{detect}, π_{decide}, π_{eval}) to non-sequential architectures requires domain-specific design decisions that have not been validated.

Threshold sensitivity. The framework uses many configurable thresholds ($\tau_{\text{sig}} = 85$, $\tau_{\text{base}} = 50$, $\tau_{\text{act}} = 85$, $\tau_{\text{comp}} = 70$, $\tau_{\text{opp}} = 0.3$, among others) whose optimal values may be behavior- and model-dependent. While percentile-based thresholds adapt automatically to model scale and training intensity, the choice of which percentile to use (e.g., 85th vs. 90th) affects the sensitivity-specificity tradeoff. Systematic sensitivity analysis across behaviors and model scales is needed.

Behavioral signal construction. The framework assumes that behavioral training signals can be cleanly constructed—that target and opposite behaviors can be operationalized as training objectives with available datasets. This holds well for studied behaviors (safety refusal, truthfulness,

sentiment) but may not hold for subtle or ill-defined behaviors where the target/opposite distinction is ambiguous or where high-quality training data is unavailable.

Peak checkpoint selection. The peak selection mechanism assumes behavioral metrics are available and well-defined during training. For behaviors where evaluation is expensive, noisy, or requires human judgment, the confirmation window ($K = 3$, $\delta = 0.05$) may not reliably identify the optimal checkpoint.

13 Conclusion

We have presented the Differential Circuit Analysis Framework (DCAF), a formally specified protocol that integrates projection-level analysis, multi-signal training perturbations, and triangulated confidence scoring into a unified pipeline for neural circuit discovery. DCAF synthesizes independently established threads in mechanistic interpretability—the QK/OV decomposition [Elhage et al., 2021], training-as-probe methodology [Prakash et al., 2024, Tigges et al., 2024], the triangulation principle [Long, 2025, Sharkey et al., 2025], and multi-method validation [Mondorf et al., 2025]—into a coherent framework with novel operational instruments: the 11-paired-objective probe configuration, the Direction Convergence Score (DCS), the 7-phase ablation protocol with interaction discovery, and adaptive tiered functional classification.

The framework is implemented as an open-source toolkit with the core mathematical framework, domain analysis modules, training infrastructure, and CLI tooling functional for transformer architectures. Several directions for future work are immediate: systematic benchmarking on MIB [Mueller et al., 2025] to enable direct comparison with existing circuit discovery methods; scaling validation to larger models (7B+) to assess whether DCAF's projection-level granularity yields discoveries that head-level methods miss; completing the direction synthesis module (DCS computation and composition analysis); and expanding architecture-specific instantiations to vision and reinforcement learning systems.

More broadly, DCAF addresses a structural gap identified by recent work: the field has spent two years documenting the limits of single-method circuit discovery [Méloux et al., 2025, Zhang and Nanda, 2024] and calling for principled multi-signal validation [Sharkey et al., 2025, Long, 2025], but has not yet supplied an integrated protocol that operationalizes these principles. DCAF is one answer to that call. Whether its specific configuration of 11 training signals, 3 discovery paths, and triangulated confidence proves to be the right integration remains an empirical question—one that the framework's formal specification and public implementation are designed to help the community answer.

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A Implementation Considerations

Baseline model selection ($\mathcal{M}_0$):

- Choose Option A ($\mathcal{M}_0 = \mathcal{M}_{\text{pre}}$) when target behavior training data does not depend on instruction-following format
- Choose Option B ($\mathcal{M}_0 = \text{IT}(\mathcal{M}_{\text{pre}}, \mathcal{D}_{\text{IT}})$) when behavioral datasets require instruction-following capabilities (safety, helpfulness, honesty, etc.)
- Option B is recommended for safety research; see Definition 3.1 for rationale

Option B validation:

- Verify $\mathcal{D}_{\text{IT}}$ is explicitly filtered to exclude target behavioral domain content
- Confirm $\mathcal{M}_0$ is a competent instruction follower (evaluate on held-out instruction benchmarks)

- Verify $\mathcal{M}_0$ does not exhibit the target behavior
- Spot-check $\mathcal{D}_{\text{IT}}$ for contamination: sample 100+ examples and verify none contain target-domain content

Computational efficiency:

- Use activation checkpointing to save memory during capture
- Parallelize signal training across GPUs
- Cache baseline model ($\mathcal{M}_0$) activations (shared across all signals)
- Precompute $\sqrt{m \cdot n}$ for each projection shape once per model

Signal-specific objectives. Each signal type defines a specific training objective:

$$\mathcal{O}_i = \begin{cases} \mathbb{E}_{(x,y)\sim\mathcal{D}_i}\left[-\log p(y \mid x)\right] & \text{if } t_i \in \{\text{SFT signals}\} \\ -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}_i}\left[M(y_w, y_l \mid x)\right] & \text{if } t_i \in \{\text{preference signals}\} \\ -\mathbb{E}_{x\sim\mathcal{D}_i}\left[R(x)\right] & \text{if } t_i \in \{\text{RL signals}\} \\ -\mathbb{E}_{x\sim\mathcal{D}_i}\,\mathbb{E}_{y\sim\pi(\cdot|x)}\left[R(x,y)\right] & \text{if } t_i \in \{\text{GRPO signals}\} \end{cases} \tag{60}$$

Anti and Negated signals negate the relevant objective (gradient ascent rather than descent).

Signal effectiveness—full formula.

$$\Delta_{\text{improve}}(i) = \begin{cases} \dfrac{\mathcal{L}_{\text{pre}}(i) - \mathcal{L}_{\text{peak}}(i)}{\mathcal{L}_{\text{pre}}(i)} & \text{SFT signals} \\ \dfrac{M_{\text{peak}}(i) - M_{\text{pre}}(i)}{|M_{\text{pre}}(i)| + \varepsilon} & \text{preference signals} \\ \dfrac{\mathbb{E}[R_{\text{peak}}(i)] - \mathbb{E}[R_{\text{pre}}(i)]}{|\mathbb{E}[R_{\text{pre}}(i)]| + \varepsilon} & \text{GRPO signals} \\ \dfrac{R_{\text{peak}}(i) - R_{\text{pre}}(i)}{|R_{\text{pre}}(i)| + \varepsilon} & \text{RL signals} \end{cases} \tag{61}$$

Normalize to $[0, 1]$ with outlier handling via clipping at 95th percentile.

Confound-specific baseline filtering. The baseline filter can be enhanced when datasets exist that specifically target known confounds:

$$\text{baseline_filter}(\text{proj}) = \prod_{t \in \mathcal{T}^0_{\text{general}} \cup \mathcal{T}^0_{\text{confounds}}} \mathbf{1}\left[\|\Delta W_t^{(\text{proj})}\|_{\text{RMS}} < \Phi_{\tau_{\text{base}}}(\cdot)\right] \tag{62}$$

Representation direction synthesis. The direction module extends per-component contrastive directions to the global residual stream, producing validated behavioral directions via DCS, layer trajectory analysis, composition decomposition, and intervention specifications. Full details of global direction extraction, consensus direction computation, dimensionality assessment, and composition analysis are specified in the complete technical specification accompanying this paper.

Parametric embedding (optional). Train a parametric function $f_\theta : \mathbb{R}^{d_k} \to \mathbb{R}^2$ to minimize PaCMAP loss, enabling embedding of new points, consistent coordinate systems across checkpoints, and generalization testing.

B Architecture-Specific Implementations

B.1 Transformer Instantiation

B.1.1 Probe Type Instantiation

For language models, the universal probe types instantiate as:

π_R **(Recognition):** Forward pass only on behavioral prompts. Captures input encoding/detection.

π_F **(Free Generation):** Generate first N tokens. Captures steering decision.

π_T **(Teacher Forcing):** Compare perplexity on target vs. opposite continuations. Captures path preference.

B.1.2 Component Decomposition

For each layer ℓ and head h:

$$Q_{\ell,h} = XW_Q^{(\ell,h)} \quad \text{(query: what to look for)} \tag{63}$$

$$K_{\ell,h} = XW_K^{(\ell,h)} \quad \text{(key: what to advertise)} \tag{64}$$

$$V_{\ell,h} = XW_V^{(\ell,h)} \quad \text{(value: what to transmit)} \tag{65}$$

$$O_{\ell,h} = \text{Attn}_{\ell,h} W_O^{(\ell,h)} \quad \text{(output: how to integrate)} \tag{66}$$

For gated MLP architectures:

$$G_\ell = \sigma(XW_{\text{gate}}^{(\ell)}) \quad \text{(gate: what passes through)} \tag{67}$$

$$U_\ell = XW_{\text{up}}^{(\ell)} \quad \text{(up: expansion to high-dim)} \tag{68}$$

$$D_\ell = (G_\ell \odot U_\ell) W_{\text{down}}^{(\ell)} \quad \text{(down: projection back)} \tag{69}$$

B.1.3 Grouped-Query Attention (GQA)

In GQA architectures, K and V projections are shared across head groups with group_size $= n_{\text{query}}/n_{\text{kv}}$. The projection-to-component mapping μ_{proj} maps each shared K/V projection to all components in its group. When a shared K/V projection shows significance, all query heads in its group are flagged.

B.1.4 Differential Analysis Methods

Differential logit lens: $\Delta L_\ell^{(i)} = L_\ell^{\text{post-}t_i} - L_\ell^{\text{pre}}$. Identifies at which layer the behavioral decision shifts.

Differential attention entropy: $\Delta H_{\ell,h}^{(i)} = H_{\ell,h}^{\text{post-}t_i} - H_{\ell,h}^{\text{pre}}$. Negative: head became more focused. Positive: more diffuse.

Differential attention patterns: $\Delta P_{\ell,h}^{(i)}[\text{final} \to B] = \sum_{b \in B}(P_{\ell,h}^{\text{post}}[n,b] - P_{\ell,h}^{\text{pre}}[n,b])$. Whether heads learned to attend to/away from behavioral content.

Differential PCA: PCA on $\Delta A = A^{\text{peak}} - A^{\text{base}}$ directly captures directions that training added.

Differential probing: Compare linear probe accuracy on baseline vs. peak-checkpoint activations.

B.1.5 Example: Safety Behavior Circuit Discovery

Target behavior: Refusing harmful requests. $\mathcal{T}^+$: SimPO/SFT on (harmful request, refusal) pairs; Anti/Negated for compliance. $\mathcal{T}^-$: SimPO/SFT on (harmful request, compliance) pairs; Anti/Negated for refusal. $\mathcal{T}^0$: General language modeling on neutral text.

Expected findings: Recognition components (early-layer attention heads detecting harmful keywords), Steering components (mid-layer MLPs implementing refusal decision), Preference components (late-layer attention heads evaluating response safety).

B.2 Vision Models (Preliminary)

For vision transformers: probe types instantiate as image embedding (π_{encode}), generation (π_{generate}), and preference scoring (π_{prefer}). Component decomposition follows the transformer pattern. For convolutional networks: convolutional kernels by layer and channel serve as components.

B.3 Diffusion Models (Preliminary)

Probe types: prompt embedding (π_{encode}), denoising trajectory (π_{denoise}), output likelihood (π_{score}). Activation capture at noise predictions, cross-attention, U-Net bottleneck, and timestep embeddings.

C Complete Notation Reference

C.1 Core Objects

Symbol	Meaning
$\mathcal{M}_0$	Configured baseline model: $\mathcal{M}_{\text{pre}}$ (Option A) or IT($\mathcal{M}_{\text{pre}}$, $\mathcal{D}_{\text{IT}}$) (Option B)
$\mathcal{M}_{\text{pre}}$	Pretrained model (before any fine-tuning)
$\mathcal{D}_{\text{IT}}$	Behaviorally neutral instruction-tuning dataset (Option B only)
$W_0 \in \mathbb{R}^d$	Parameters of $\mathcal{M}_0$, d = total param count
$\mathcal{K} = \{k_1, \ldots, k_m\}$	Components (graph nodes): attention heads, MLP layers
$\mathcal{P}$	All projections: $W_Q, W_K, W_V, W_O, W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}$
$\mu_{\text{proj}} : \mathcal{P} \to \mathcal{K}$	Maps each projection to its containing component

C.2 Training Signals and Checkpoints

Symbol	Meaning
$\mathcal{T}$	All 11 training signals: $\mathcal{T}^+ \cup \mathcal{T}^- \cup \mathcal{T}^0$
$\mathcal{T}^+$	Target behavior cluster (5 signals)
$\mathcal{T}^-$	Opposite behavior cluster (5 signals)
$\mathcal{T}^0$	Neutral baseline (1 signal)
t_1–t_5	PrefOpt, SFT, Cumulative, Anti, Negated for target
t_6–t_{10}	Mirror of t_1–t_5 for opposite behavior
t_{11}	DomainNative: neutral task
$W_{i,\text{peak}}$	Peak checkpoint: confirmed-best behavioral performance
$\text{eff}(i)$	Normalized signal effectiveness $\in [0, 1]$

C.3 Multi-Path Discovery

Symbol	Meaning
$\mathcal{H}_W, \mathcal{H}_A, \mathcal{H}_G$	Weight-, activation-, gradient-discovered projection sets
$\mathcal{H}_{\text{disc}}$	All discovered components
$S_W^{(\text{proj})}, S_A^{(\text{proj})}, g^{(\text{proj})}$	Projection-level discovery scores
$\text{paths}(k)$	Number of discovery paths that flagged component k
$\text{bonus}(k)$	Multi-path bonus: $\beta_{\text{path}} \cdot \max(0, \text{paths}(k) - 1)$

C.4 Weight Domain

Symbol	Meaning
$\Delta W_i^{(\text{proj})}$	Weight delta: $W_{i,\text{peak}}^{(\text{proj})} - W_0^{(\text{proj})}$
$\lVert \cdot \rVert_{\text{RMS}}$	RMS-normalized Frobenius: $\lVert \cdot \rVert_F / \sqrt{m \cdot n}$
$\bar{\delta}_+^{(\text{proj})}, \bar{\delta}_-^{(\text{proj})}$	Effectiveness-weighted cluster delta matrices
$\text{opp_degree}(\text{proj})$	Opposition: $\max(0, -\text{cos_opp})$
$\text{sig}(\text{proj}, i)$	Binary significance predicate
$C_W^{(\text{proj})}, C_W^{(k)}$	Projection- and component-level weight confidence

C.5 Activation Domain

Symbol	Meaning
$\Pi = \{\pi_{\text{detect}}, \pi_{\text{decide}}, \pi_{\text{eval}}\}$	Probe types
$A_i^{(k,\pi)}$	Activations at component k, probe π, signal i
$m_i^{(k,\pi)}$	Activation magnitude
$C_A^{(k)}$	Activation confidence

C.6 Geometry Domain

Symbol	Meaning
$d_i^{(k)}$	Contrastive direction at component k, signal i
$\text{coh}_+, \text{coh}_-, \text{opp}, \text{orth}$	LRS alignment components
$\text{LRS}^{(k)}$	Linear Representation Score
$C_G^{(k)}$	Geometric confidence: LRS $\times$ generalization
$\text{DCS}(L)$	Direction Convergence Score at layer L
$d_{\text{consensus}}^{(L)}$	Consensus direction at layer L
L^*	Peak DCS layer

C.7 Confidence and Classification

Symbol	Meaning
$C_{\text{base}}^{(k)}$	Base triangulated confidence
$C^{(k)}$	Unified confidence: $\min(1, C_{\text{base}} + \text{bonus})$
$\mathcal{H}_{\text{cand}}, \mathcal{H}_{\text{conf}}$	Candidate and confirmed component sets
$G = (V, E)$	Behavioral circuit graph
$v_{k,+}^*, v_{k,-}^*$	Bidirectional steering vectors
$\alpha_+, \alpha_-, \alpha_\pm$	Steering-geometry alignment scores
$I_\pi^{(k)}$	Ablation impact under probe π
$P(k), A(k)$	Primary and auxiliary functional tiers

D Threshold Parameters and Defaults

Discovery thresholds:

Parameter	Default	Description
τ_{sig}	85	Weight significance percentile
τ_{base}	50	Baseline threshold percentile
τ_{act}	85	Activation significance percentile
τ_{grad}	85	Gradient discovery threshold percentile
τ_{comp}	70	Component screening threshold percentile
β_{path}	0.15	Multi-path discovery bonus weight

Weight domain:

Parameter	Default	Description
α	0.2	Opposition bonus weight
τ_{opp}	0.3	Bidirectionality threshold
β	0.4	Threshold bonus weight
q	2	Effectiveness power for confidence weighting

Geometry and LRS:

Parameter	Default	Description
p	0.5	LRS / DCS power mean parameter
ε	0.05	Smoothing parameter
$\tau_{\text{nonlinear}}$	0.4	LRS threshold for nonlinear diagnostics

Direction synthesis:

Parameter	Default	Description
$\tau_{\text{dcs-conv}}$	0.3	DCS convergence prerequisite minimum
$\tau_{\text{dcs-peak}}$	0.5	DCS emergence/peak detection
τ_{stab}	0.9	Direction stability for stable range
τ_{fidelity}	0.8	Composition reconstruction fidelity

Confidence and candidate selection:

Parameter	Default	Description
τ_{unified}	0.3	Unified confidence threshold
w	2	Weight domain emphasis in triangulated confidence
τ_{orphan}	0.6	High-confidence orphan threshold

Ablation and classification:

Parameter	Default	Description
$r_{\text{primary}}^{\text{strict}}$	0.8	Strict primary function threshold
$r_{\text{primary}}^{\text{relaxed}}$	0.7	Relaxed primary function threshold
r_{aux}	0.6	Auxiliary function threshold
τ_{abs}	0.15	Absolute minimum impact
τ_{gap}	0.10	Tight clustering gap threshold
ε_{syn}	0.05	Synergy detection threshold
B_{pair}	300	Maximum pairwise ablation tests (Phase 2)
B_{triple}	100	Maximum triple ablation tests (Phase 5)

Steering and peak checkpoint:

Parameter	Default	Description
τ_{ppl}	0.1	Max perplexity increase (steering)
K	3	Peak confirmation window
δ	0.05	Peak stability tolerance